

Mathematics Standard 2

HSC marking feedback 2025

General feedback

Students should:

- show relevant mathematical reasoning and/or calculations
- read the question carefully to ensure they do not miss important components of the question
- have a clear understanding of key words in the question and recognise the intent of the question such as show, solve, evaluate, hence, calculate
- use the Reference Sheet where appropriate
- ensure the solution is legible and follows a clear sequence
- engage with any stimulus material provided and refer to it in their response when required by the question
- check their solution answers the question
- round off numerical solutions only at the final step of the solution
- construct graphs neatly, with precision, and display all relevant information as required by the question
- interpret information presented in graphs across a range of contexts
- understand when to use relevant calculator functions
- note carefully any information in the questions which supplies units of measurement.

Section II

Question 16

In better responses, students were able to:

- calculate the cost of buying 80 litres of paint in 10-litre cans correctly
- calculate the cost of buying 80 litres of paint in 4-litre cans correctly
- calculate the savings by finding the difference between the cost of 10-litre and 4-litre cans.

Areas for students to improve include:

- showing the difference of two calculated values
- calculating the amount saved for the whole amount of 80 litres, not just 1 litre.

Question 17

In better responses, students were able to:

- plot 2 points correctly and draw a line of best fit through the points using a ruler
- determine the y -intercept and gradient from the diagram
- recognise a negative gradient slopes downward from left to right
- write the equation of straight line correctly in the form $y = mx + c$
- recognise that the point $(0,14)$ gives the y -intercept of the line of best fit.

Areas for students to improve include:

- determining if the gradient is positive or negative
- demonstrating understanding that $y = mx + c$ and understanding what m and c represent
- recognising the x -value and y -value in a coordinate point to plot them on a Cartesian plane.

Question 18

In better responses, students were able to:

- identify the correct interest factor from the table
- use the future value interest factor from the table to find the value of the prize at the end of 10 years.

Areas for students to improve include:

- leaving the calculations incomplete through not multiplying by the number of years, that is, 10
- using the table provided and not a formula
- demonstrating understanding of the purpose of the table to assist with the calculation of the annuity.

Question 19

In better responses, students were able to:

- determine the immediate prerequisites of activities from the network diagram and transfer information to the table
- identify the critical path and write it down
- state the minimum duration of the project
- apply forward and backward scanning to find Earliest Starting Time and Latest Starting Time.

Areas for students to improve include:

- identifying an immediate prerequisite
- demonstrating understanding of a critical path
- demonstrating understanding of the Earliest Starting Time and Latest Starting Time and how to calculate them
- demonstrating understanding that the completion time of a project is calculated by finding the highest duration value of each activity, and that an increase in duration may or may not affect the critical path.

Question 20 (a)

In better responses, students were able to:

- find t using the axis of symmetry
- substitute the value of t into the equation of the quadratic to find h .

Areas for students to improve include:

- recognising the difference between intercepts and the vertex.

Question 20 (b)

In better responses, students were able to:

- use symmetry to find the value of t
- substitute $h = 12$ and correctly solve for t .

Areas for students to improve include:

- recognising the symmetrical shape of a quadratic
- solving a quadratic by factorising.

Question 21

In better responses, students were able to:

- write an equation and solve to find the number of hours
- give a clear, labelled expression for the cost of electricity used
- convert cents into dollars before calculation.

Areas for students to improve include:

- recognising the requirement for a specific cooling or heating cost/time period, not a combined one
- providing all steps required, rather than just dividing total cost by number of days in calculating the number of hours the air conditioner was used.

Question 22

In better responses, students were able to:

- include only edges on the cut flowing from the source to the sink
- calculate the capacity of the indicated cut
- identify an actual path
- connect the flow network to the original network diagram in part (a).

Areas for students to improve include:

- demonstrating understanding that the capacity of the cut only includes flows from the source towards the sink
- showing the inflow versus outflow at the source and sink on the diagram
- investigating different ways a flow network can flow efficiently, with the maximum flow and less than the maximum flow
- demonstrating understanding of the difference between a flow network and a critical path network.

Question 23

In better responses, students were able to:

- calculate yields for both companies and compare results accurately
- interpret that a higher dividend yield represents a better investment
- demonstrate a solid understanding of dividend, share price, and yield terminology.

Areas for students to improve include:

- writing clear concluding statements and interpreting results meaningfully.

Question 24

In better responses, students were able to:

- write the ratios for capture-recapture
- equate the two populations and tagged snails correctly
- identify the need to work backwards to find the second sample of tagged snails.

Areas for students to improve include:

- knowing that multiplying a fraction by the denominator of another fraction, that denominator is transposed to the numerator on the other side of the equation
- using fractions to correctly represent the ratios.

Question 25 (a)

In better responses, students were able to:

- demonstrate a strong understanding through accurately selecting the correct terminology to describe both form and direction
- identify that the form is linear and the distribution has a negative direction.

Areas for students to improve include:

- using the correct language for form and/or direction
- refining their knowledge of statistical terms and key words used in the syllabus for bivariate data analysis.

Question 25 (b)

In better responses, students were able to:

- identify the slope and y -intercept for the equation of the least-squares regression line
- clearly interpret the meaning of the slope and y -intercept within the context of the scenario, rather than simply explaining what slope and y -intercepts represent in general terms.

Areas for students to improve include:

- interpreting the slope and y -intercept in context of the question with values shown
- interpreting the slope and y -intercept correctly, not just identifying the values
- understanding that the slope is not the same as the correlation coefficient.

Question 25 (c)

In better responses, students were able to:

- correctly substitute $x = 42$ into the equation to calculate the answer.

Areas for students to improve include:

- showing correct substitution into the formula
- understanding what each variable in an equation represents.

Question 25 (d)

In better responses, students were able to:

- convert 2 hours to minutes and substitute $x = 120$ minutes into the formula
- identify that the graph goes below zero and cannot have a negative number of minutes of exercise
- recognise 2 hours is outside the dataset
- understand the meaning of the key word 'extrapolate'.

Areas for students to improve include:

- understanding that the values outside the dataset are not reliable
- knowing extending the line of regression gives the negative answer for time
- knowing the meaning of the key word extrapolate.

Question 26

In better responses, students were able to:

- apply the trapezoidal rule correctly, identifying the height, that is, $h = 5.1$, and appropriate parallel sides
- use two applications of the trapezoidal rule where required
- apply absolute and percentage error formulas accurately
- convert between units, that is, $\text{mm} \leftrightarrow \text{cm}$, and round to the correct number of significant figures.

Areas for students to improve include:

- applying and rounding to a specific number of significant figures consistently
- demonstrating understanding of absolute and percentage error and calculating these correctly
- converting units correctly and ensuring consistent use.

Question 27

In better responses, students were able to:

- calculate the number of days correctly
- use the compound interest formula correctly to calculate the amount owed.

Areas for students to improve include:

- calculating the interest per day as a percentage or decimal

converting the interest rate and time periods to the correct unit.

Question 28

In better responses, students were able to:

- mark the quartiles clearly on the box-plot
- provide the five-number summary.

Areas for students to improve include:

- using the graph to demonstrate where the quartiles lie
- interpreting and finding values from a cumulative frequency histogram and ogive
- demonstrating understanding of what a box-plot looks like.

Question 29

In better responses, students were able to:

- draw a probability tree diagram without replacement
- multiply along the branches correctly and add both options
- write the correct fractions along the branches clearly.

Areas for students to improve include:

- demonstrating understanding of 'without replacement' in probability scenarios
- using a tree diagram to show the selections
- demonstrating understanding of when to add or multiply probabilities.

Question 30

In better responses, students were able to:

- provide two equations correctly using the pronumerals given in the question
- interpret the information to retrieve correct pronumerals.

Areas for students to improve include:

- demonstrating understanding of the need to write two equations
- demonstrating understanding of the format of linear equations.

Question 31

In better responses, students were able to:

- work with order of operations to understand at what point to divide by 0.37
- choose the correct tax bracket
- find the additional tax payable.

Areas for students to improve include:

- making sure they stay in the same bracket for all the calculations
- converting 37 cents to 0.37 before doing any calculations
- demonstrating understanding of the difference between tax payable and taxable income.

Question 32

In better responses, students were able to:

- find the volume of a sphere and a pyramid
- interpret that the remaining space needs to be found
- multiply the volume of the sphere by 30 and then add the given volume to go on to find h .

Areas for students to improve include:

- knowing the formulas to find the volumes of a pyramid and a sphere
- using given information, that is, the number of spheres placed inside the pyramid, to support calculations, rather than relying on the diagram to estimate.

Question 33

In better responses, students were able to:

- calculate the total stamp duty
- calculate the interest on the total borrowed, not just on the cost of the car
- find the total owing correctly, and understand it was the total borrowed, plus the interest.

Areas for students to improve include:

- calculating the stamp duty on the sale price of the car
- demonstrating understanding of what is meant by a monthly repayment.

Question 34

In better responses, students were able to:

- calculate $n = 84$, $r = \frac{6}{12}\% = 0.005$ and identify from the future value table, the interest factor of 104.07393
- use the future value formula correctly: $A = 21000 \left(1 + \frac{6}{12}\%\right)^{84}$
- calculate the monthly deposit correctly: $A \div 104.07393 = \$306.78$.

Areas for students to improve include:

- converting the interest rate (r) and number of periods (n) to the compounding period
- recognising when to use the future value formula and when to use the provided future value interest factors tables
- understanding when to divide by the interest factor and when to multiply.

Question 35

In better responses, students were able to:

- identify the angle of depression on diagrams correctly and understand its relationship to the angle of elevation
- recognise when non-right-angled trigonometry is required
- apply sine and cosine rules accurately, showing clear substitution and working
- demonstrate understanding of alternate and corresponding angles in parallel lines
- recognise and handle the ambiguous case in the sine rule.

Areas for students to improve include:

- placing and interpreting angles of depression on diagrams correctly
- demonstrating understanding of when to use sine or cosine rules, and the information required for each
- recognising ambiguous sine rule cases.

Question 36

In better responses, students were able to:

- use the formula to find the salvage values at the end of the 9th and 10th years
- calculate the amount the car depreciates in the 10th year
- demonstrate understanding that the 10th year is not at the end of year number 10, but how much the car has dropped in value during that 10th year.

Areas for students to improve include:

- demonstrating understanding that depreciating means the value is decreasing as time progresses
- demonstrating understanding that with a declining balance, the depreciating amount is not a constant value but rather a rate/percentage
- working out a rate using a graph.

Question 37

In better responses, students were able to:

- use the area formula for non-right-angled triangles
- label the new information, then calculate on the diagram
- recognise that the shaded grassed area = the area of the bigger equilateral triangle – 3 times the area of one of the small triangles
- use ratios correctly to find the time taken to mow the area.

Areas for students to improve include:

- knowing when to use the relevant trigonometric formulas to calculate area and sides of both right-angled and non-right-angled triangles
- demonstrating understanding of geometric properties to find unknown angles within the triangles
- using ratios as applied to the problem presented.

Question 38

In better responses, students were able to:

- convert between different units, including unfamiliar units
- convert miles per gallon to 3.8 L/48 km
- complete distance and volume conversion calculations without mixing them up.

Areas for students to improve include:

- completing one conversion at a time, rather than attempting multiple at once
- using words as a support strategy to connect to their calculations to establish next steps on further calculations and conversions.

Question 39

In better responses, students were able to:

- recognise an exponential equation and its key features on a graph
- substitute values for (x, y) and/or k correctly to attempt to find the value of y when $x = 4$
- apply key features of the exponential graph, for example, y -intercept, to determine the equation.

Areas for students to improve include:

- distinguishing between different types of graphs and functions
- demonstrating understanding of the link between the y -intercept and k .

Question 40 (a)

In better responses, students were able to:

- find the number of male sheep correctly using the given information: $\frac{12600}{21} = 600$
- express 15 male sheep as a percentage of 600 male sheep: $\frac{15}{600} = 0.025 = 2.5\%$
- use the percentage to equate to a z -score of 2
- calculate x kg using the normal distribution curve or the z -score formula.

Areas for students to improve include:

- calculating the number of male sheep correctly according to the given ratio
- identifying the difference between a z -score and a percentage
- using and labelling a diagram to help to understand where the values are placed within the normal distribution
- using the Reference Sheet to match the percentage break-up to a z -score.

Question 40 (b)

In better responses, students were able to:

- express 300 female sheep as a percentage of the 12 000 female sheep correctly
- recognise the female percentage (or z -score) is the same as the male percentage from part (a) of the question
- recognise and state that with a smaller mean and standard deviation, the percentage of female sheep greater than x would be less than 2.5%, or that the z -score of x would be greater than 2.

Areas for students to improve include:

- calculating the correct number and percentage of female sheep
- understanding the concepts, z -score, mean, standard deviation in more detail, and how these concepts affect each other
- providing an accurate explanation, justified with mathematical reasoning and calculations.